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SECTOR AT A GLANCE 05

Watershed Management in India

Water in India

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WATERSHED MANAGEMENT IN INDIA

WATER IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Watershed management is the coordinated approach to managing the resources of a particular drainage basin for improved water quality and supply, and sustainable land use. It encompasses soil conservation, water conservation, and agricultural practices tailored to the unique characteristics of the watershed area.¹ Done well, watershed management can improve water availability and enhance agricultural productivity, climate resilience, biodiversity conservation, and aid sustainable rural development.²

Watershed management, thus, assumes special importance in India given how critical agriculture and water resources are for food security and rural livelihoods. Scientific watershed management is also important in India since 37% of the land area in the country is facing degradation³, water scarcity is a growing concern, with 63% of Indian districts experiencing depleting groundwater levels⁴, and India has been assessed as the 7th most vulnerable country to extreme climate change events.⁵

Given this background, watershed management has been a key focus in recent years. Several government initiatives, including but not limited to the Watershed Development Component (WDC) of the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), and a range of States such as Rajasthan, Andhra Pradesh, Maharashtra, Kerala, and several more focus on sustainable watersheds, with the eventual goal of ensuring that India achieves food, ecological, and economic security.

Progress has been made thanks to these efforts resulting in the creation of over 6,000 watershed projects and 330,000 micro-watersheds under PMKSY. Similarly, 0.6 million water harvesting structures have been created since 2014-15.

However, there remain significant challenges. 30% of India's geographic area is degraded due to erosion, salinity, alkalinity, or vegetation loss. Citizen satisfaction with the watershed component of PMKSY is low. Addressing these and other similar challenges will be critical going forward and will require ongoing commitment and adaptation based on local contexts.

This note describes key trends across various sub-components of watershed management: land use management, watershed conservation, soil conservation, afforestation, and community engagement. It also describes key challenges and opportunities, and some ideas for the way forward.

Figure 1: Key watershed management statistics



Source: NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

¹ World Resources Institute, India's Watershed Development Boosts Food Security, Improves Livelihoods, 2013

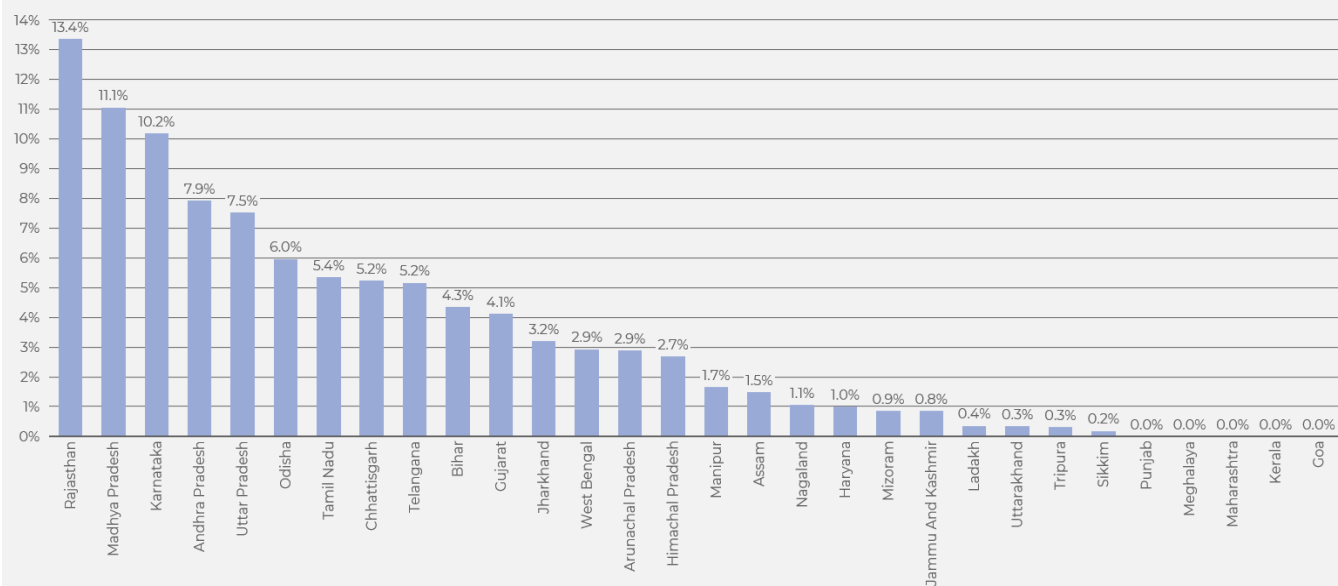
² World Resources Institute, India's Watershed Development Boosts Food Security, Improves Livelihoods, 2013

³ National Academy of Agricultural Sciences, Achieving Land Degradation Neutrality in India, 2021

⁴ NITI Aayog, CWMI 1.0 & CWMI 2.0, 2018/19; World Bank, India seeks to arrest its alarming decline in groundwater, 2022

⁵ CEEW, Mapping India's Climate Vulnerability, 2021

Figure 2: Distribution of micro-watersheds across states/UTs (2023)



Source: Ministry of Rural Development, State-wise Number of Watersheds, 2023

Key trends

This section dives deeper into various components of watershed management in India:

1. Land Use Management
2. Soil Conservation
3. Water Conservation
4. Afforestation
5. Community Engagement

Land Use Management

Land use management refers to policies and practices that how land and its resources are used, developed, and conserved within a watershed. Some aspects include zoning and land use planning, conservation easements, low or zero tillage for conservation, wetland restoration, as well as regular monitoring.

High rates of land degradation make land use management critical, especially for the welfare of smallholder farmers:

Around 30% of the total geographical area of India is degraded due to erosion, vegetation loss, salinity, and alkalinity.⁶ The effects of such degradation are more severe in rainfed areas, which are vulnerable to climate change and bring risks for 80% of the nation's small and marginal farmers who depend on them.⁷

Soil Conservation

Watershed management schemes in India aim to prevent soil run-off and, to harness, conserve, and develop natural resources, including soil.⁸ Initiatives such as staggered trenching, graded bunding, bench terracing, and contour bunding, are aimed at soil and moisture conservation efforts.⁹

Soil erosion negatively affects agriculture and the country's economy:

Soil erosion in India leads to an annual loss of 5,334 million tonnes of soil due to insufficiently judicious use of fertilisers, pesticides, and insecticides.¹⁰ The economic costs of land degradation and land use change more broadly are significant, and have been estimated at 2.5% of India's GDP in 2014-15 and almost 16% of India's gross value added from the agriculture, forestry, and fishing sectors.¹¹

Figure 3: State-wise cultivable area affected by soil erosion in India (2022)

6 NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

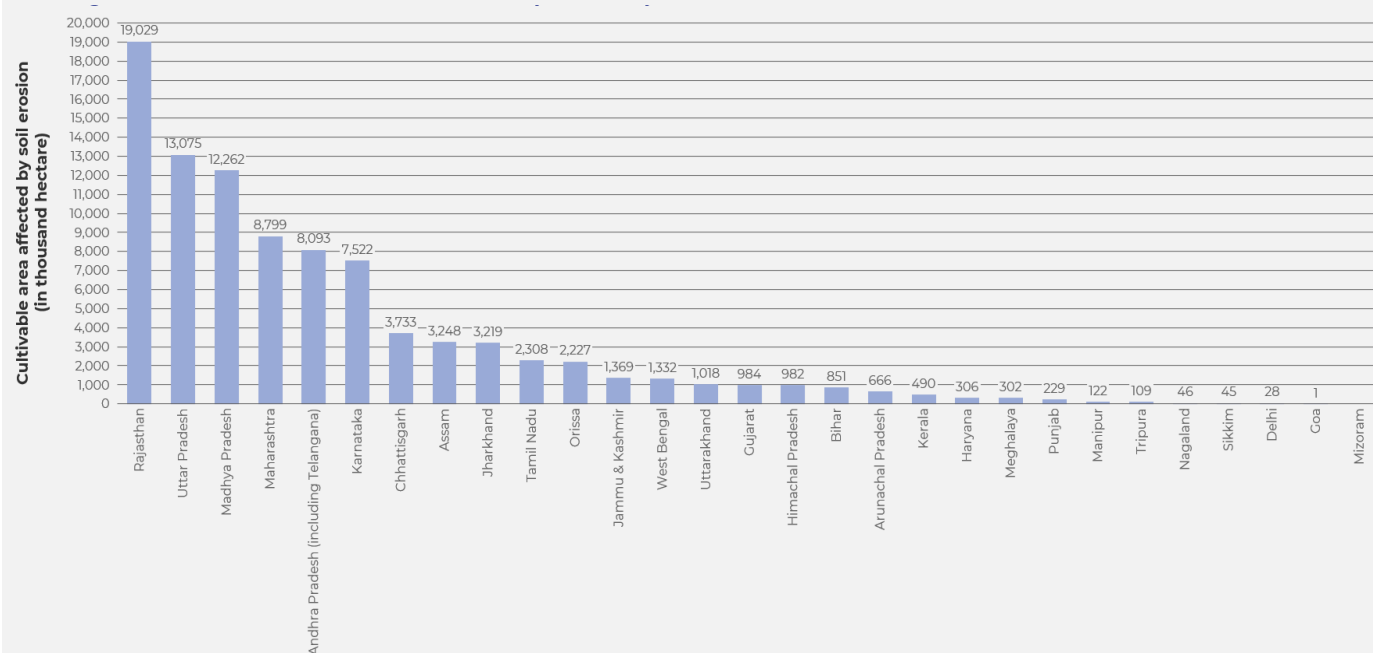
7 NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

8 NITI Aayog, Micro Irrigation Final EvaluationReport.pdf, 2020

9 Department of Land Resources, Physical Achievement Report of a State/District up to a reported month, accessed June, 2023

10 PIB, The links between Soil Fertility and Poverty alleviation, 2016

11 TERI, Economics of Desertification, Land Degradation and Drought in India, 2018



Source: Ministry of Agriculture & Farmers Welfare, Survey on Soil Erosion, 2022

Vegetation degradation is the leading cause of soil erosion in India:

Vegetation plays a crucial role in safeguarding soil and its fertility. According to the Desertification and Land Degradation Atlas of India, vegetation degradation, which involves a decrease in biomass and ground cover, often due to deforestation or overgrazing, has significantly contributed to soil erosion and the decline in soil organic content in India.¹²

Beneficiaries in watershed projects report mixed success of interventions in containing soil erosion:

According to a household survey, approximately 25% of respondents mentioned there was a decrease in soil erosion due to the watershed project. However, ~18% reported an increase in soil erosion compared to the conditions before the project.¹³

Water Conservation

Water conservation, through rainwater harvesting and recharge of the groundwater table, is a key objective of watershed management in India.¹⁴ This involves the development of structures such as farm ponds, check dams, nallah bunds, percolation tanks, and other groundwater recharge infrastructure.¹⁵

Water scarcity in rainfed areas can cause low agricultural productivity:

Rainfed areas today face a significant challenge of water scarcity, resulting in low agricultural productivity of only 1-1.5 tons per hectare.¹⁶

Watershed projects have effectively been able to recharge surface and groundwater in such regions:

An increase in surface and groundwater has been witnessed in almost all the micro watershed projects established. For instance, in Tamil Nadu, recharge shafts were constructed to recharge shallow aquifers,

¹² ISRO, Desertification and Land Degradation Atlas of India, 2021

¹³ NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

¹⁴ NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

¹⁵ Department of Land Resources, Physical Achievement Report of a State/District up to a reported month, accessed June 2023

¹⁶ ICRISAT, Pradhan Mantri Krishi Sinchai Yojana: Enhancing impact through demand driven innovations, 2016

and consequently, a rise in groundwater level from 0.33 meters to 3.28 meters was witnessed in watershed areas.¹⁷

Afforestation

Increasing the area under forests is one of the key methods to develop land under watershed development projects.

Forest cover in India has stabilized and has been increasing constantly¹⁸:

In 2021, forest cover in India was ~70 million ha, accounting for 21.71% of the nation's geographical area.¹⁹

Key watershed schemes have assisted afforestation efforts in the country:

Till March 2023, the total area brought under afforestation via the WDC (and the erstwhile Integrated Watershed Management Program) was ~600,000 ha, of which ~100,000 ha has been to cover wastelands.²⁰

Community Engagement

Guidelines for watershed management recognise community participation as a guiding principle, highlighting the need for their involvement in planning, budgeting, implementation, and management of watershed projects.²¹ Consequently, the WDC within PMKSY emphasizes the strengthening of community-based local institutions for livelihood promotion and watershed sustainability as a key objective.²²

Watershed management projects enhance livelihood generation in watershed areas:

A large share of MNREGA's water and land management projects are implemented in convergence with the PMKSY's watershed component.²³ Research suggests that the mean additional employment generation per annum in watershed areas from various activities and watershed operations ranges from 181 to 900 person-days per hectare per year.²⁴

Community satisfaction with information and support provided under the WDC component of the PMSKY can be improved:

~40% of respondents to a survey on WDC PMKSY reported that they were dissatisfied with the information provided about the scheme.²⁵ Moreover, only ~26% and ~33% of respondents were satisfied with the financial support and equipment provided under WDC PMKSY, respectively.²⁶

17 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

18 Ministry of Environment, Forest, and Climate Change, PIB, Increased in forest cover, 2022

19 Ministry of Environment, Forest, and Climate Change, PIB, Increased in forest cover, 2022

20 Department of Land Resources, Physical Achievement Report of a State/District up to a reported month, accessed June 2023; Ministry of Environment, Forest, and Climate Change, PIB, Increased in forest cover, 2022

21 GoI, Common Guidelines for Watershed Development Projects 2008, 2011

22 Department of Land Resources, Physical Achievement Report of a State/District up to a reported month, accessed June 2023

23 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

24 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

25 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

26 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective watershed management practices. These revolve primarily around the potential for improved agricultural productivity and better economic outcomes for farmers.



Improvement in agricultural productivity in rainfed areas:

Presently, 49% of India's net sown area is irrigated, and the remaining (51%) is rainfed.²⁷ On achieving full irrigation potential in the country, an additional 11% will become irrigated with higher agricultural productivity, and 40% will remain rainfed.²⁸ Since land productivity in rainfed areas is relatively poor compared to canal irrigation (2-3x) and well-irrigated areas (3-5x), this can aggravate food security challenges for the country.²⁹ Considering this, watershed development programs can be valuable instruments for developing rainfed areas and enhancing agriculture productivity in India.³⁰



Improvement in economic outcomes for farmers:

80% of small and marginal farmers most vulnerable to monsoon failures depend on rain-fed areas for their livelihood.³¹ Watershed programs, by bringing irrigation facilities to these areas and developing water harvesting structures, can improve economic outcomes for farmers, increase their climate resilience and reduce their debt burdens.³²



Improvement in rural livelihoods and incomes:

A large share of Mahatma Gandhi National Rural Employment Guarantee Act's (MNREGA) water and land management projects are implemented in convergence with the PMKSY's watershed component.³³ Research suggests that additional employment generation per annum in watershed areas from various activities, including construction, and watershed operations ranges from 181 to 900 person-days per hectare per year.³⁴ Hence, watershed investments can support poverty alleviation by enhancing employment in rural areas, which can strengthen beneficiaries' purchasing power and reduce household income disparities.³⁵

Challenges

This section highlights some of the major challenges and constraints facing watershed management in India. Efforts to resolve such challenges will ensure more efficient implementation and sustainability of watershed projects and greater citizen involvement.



Inefficient land sanction rates and classification of land resources:

Till March 2023, only ~43% of the available treatable area has been sanctioned for watershed projects under watershed-related programs.³⁶

²⁷ Department of Agriculture & Farmers Welfare, Rainfed Farming Systems, accessed July 2023; NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

²⁸ NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

²⁹ NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

³⁰ NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

³¹ NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020

³² Centre for Rural Studies, Impact and Effectiveness of "Watershed Development Programmes" in India, 2010; NITI Aayog, Evaluation of Centrally Sponsored Schemes in Water Resources, Environment and Forest Sector, 2020; ICRISAT, Pradhan Mantri Krishi Sinchai Yojana: Enhancing Impact through Demand Driven Innovations, 2016

³³ NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

³⁴ NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

³⁵ NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

³⁶ Department of Land Resources, Treatable area in state, accessed June 2023 (Note: the dataset (checked till March 2023) provides information on (1) untreatable area, (2) area covered in pre-IWMP projects, (3) area covered in other watershed programs, (4) area sanctioned under IWMP till date, (5) area with assured irrigation, and (6) net-treatable area. To calculate the percentage above, a simple division has been performed, taking (6) as the numerator and the sum of (2), (3), (4), (5), (6) as the denominator)



Limited sustainability post-implementation of watershed projects:

Current guidelines do not have adequate provisions to ensure the long-term sustainability of natural resources, and some watershed projects have encountered issues related to unregulated groundwater extraction and a shift to water-intensive crops following implementation.^{37,38} Moreover, the maintenance of structures post-implementation necessitates a user-driven Watershed Development Fund, to which a significant proportion of users do not contribute.



Inadequate afforestation efforts and use of scientific reforestation methods:

As of March 2023, only ~52% of the total targeted area (i.e., 1.16 million hectares) was brought under afforestation.³⁹ Moreover, despite reforestation efforts being underway, the impact of simply planting trees may have unintended consequences.⁴⁰ For instance, evapotranspiration rates from trees are significantly higher compared to other vegetation types, potentially leading to a reduction in local water availability.⁴¹



Limited participation and capacity-building of citizens:

~61% of people surveyed during the impact evaluation of WDC PMKSY reported that no consultation meeting with farmers was organized under the program, as against 22% who acknowledged the organization of such interactions.⁴² Along similar lines, ~64% of respondents to a survey on WDC reported that they had not attended any training for operation and maintenance under the scheme, while ~26% reported no such training had been organized at all.⁴³

37 NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

38 NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

39 Department of Land Resources, Physical Achievement Report of a State/District up to a reported month, accessed June 2023

40 World Bank, Watershed Development in India, 2014

41 World Bank, Watershed Development in India, 2014, (Note: Evapotranspiration is essentially the loss of water in form of vapor to the atmosphere by a combined process of surface water evaporation and transpiration from plants)

42 NITI Aayog, Micro Irrigation Final Evaluation Report.pdf, 2020

43 NITI Aayog, Micro Irrigation FinalEvaluationReport.pdf, 2020

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee watershed management.



Ministry of Jal Shakti

The ministry aims to conserve water resources in India, ensuring equitable access to safe and clean water while promoting sustainable water practices across various sectors.



Ministry of Agriculture & Farmers' Welfare

The ministry formulates and implements policies and programs to enhance irrigation infrastructure, promote efficient water use in agriculture, develop rainfed areas, and support farmers in adopting sustainable water management practices for improved agricultural productivity.



Ministry of Rural Development

The ministry formulates and implements policies and programs like the Integrated Watershed Management Program (IWMP) to facilitate watershed development in India. These initiatives aim to conserve soil and water, promote afforestation, and enhance the overall socio-economic conditions of rural communities.



Ministry of Environment, Forests and Climate Change

The Ministry of Environment, Forests and Climate Change in India initiates afforestation programs strategically targeting watershed areas, planting trees to prevent soil erosion and regulate water flow, thereby contributing significantly to watershed development and sustainable water resource management.



National Rainfed Area Authority

The authority is responsible for identifying rainfed regions across various States and formulating watershed development plans that integrate natural resource management.

Major policies and programs

This section highlights existing policies related to watershed management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Watershed Development Component under Pradhan Mantri Krishi Sinchayee Yojana (WDC-PMKSY)⁴⁴

Watershed development is a component under the PMKSY scheme which aims to improve the productive potential of rainfed/ degraded land through integrated

⁴⁴ Ministry of Rural Development, Watershed Development Component of PMKSY 2.0, accessed December 2023

watershed management. The Department of Land Resources (DoLR) sanctions projects to states/UT covering land targets under WDC-PMKSY 2.0 and Central grants are released.



Rejuvenating Watersheds for Agricultural Resilience through Innovative Development (REWARD) program⁴⁵

This World Bank-assisted watershed development programme aims to “strengthen capacities of National and State institutions to adopt improved watershed management for increasing farmers resilience and support value chains in selected watersheds of participating states”. The Centre will be responsible for the management, monitoring, communication and knowledge-sharing functions carried out by DoLR. Further, at the State level, the REWARD program will be contiguous to the scope of WDC-PMKSY 2.0 and will support the implementation of key science-based activities.

⁴⁵ Ministry of Rural Development, Ajay Tirkey, Secretary, DoLR reviewed REWARD Program with World Bank Team, 2023

Way forward

Continued focus on watershed management is going to be critical to India's food and water security going forward. Progress is being made through programs such as WDC-PMKSY and a strong foundation has been put in place through a large number of micro-watersheds being developed. However, there are still challenges related to widespread participation, sustainability considerations, and even technical aspects of these projects.

Going forward, a focus on four themes could accelerate progress:

 Adopting a systems lens	• Taking a systems view that looks not only at micro-watersheds but also at how these align with overall sub-basin and basin-level strategies to manage water and land resources will necessitate getting scientific expertise and also a wide spectrum of stakeholders into planning.
 Planning for sustainability	• Ongoing O&M of these watershed structures will be critical. Provisions should be made towards ongoing financial support for this O&M both through government and community contributions. Innovative financing instruments, such as social impact bonds, may be tried out as well to crowd in the private sector and philanthropic capital for watershed management.
 Equipping communities with better tools	• Recent years have seen significant progress in terms of data, analysis, and watershed planning tools that enable communities to “see” the local watershed and water ecosystem around them and plan their actions accordingly. These tools should be deployed at scale to enable better watershed planning that is grounded in local needs and realities.
 Equity and behaviour change	• Watershed management is a deeply human activity and care must be taken that equity considerations are kept front in center so that all rural smallholders, both upstream and downstream benefit. Similarly, care must be taken to build behaviour change and buy-in for these programs.

Recommended readings

NITI Aayog. (2021). *Compendium of Best Practices in Water Management 2.0*.

https://www.niti.gov.in/sites/default/files/2021-11/Compendium-of-Best-Practices-in-Water-Management_03-11-2021_compressed.pdf

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